6COSC023W

Computer Science Final Project

Project Proposal and Requirements Specification (PPRS)

Handout

Misinformation Detection Software for X (formerly Twitter)

Student: Victor Okafor(w1832950)

Supervisor: Mohammed Sha Mohammed Kunju

Degree: BEng (Hons) Software Engineering

School of Computer Science & Engineering

College of Design, Creative and Digital Industries

University of Westminster

13/11/2025

Declaration

This document has been prepared based on my own work. Where other published and unpublished source materials have been used, these have been acknowledged in references.

Student Name: Victor Okafor

Date of Submission: 13/11/2025

1. Definition of the Problem

- **Problem Description**:

Misinformation is a massive detriment to a society, especially in our current digital age, the massive use and impact of social media and communication technologies has exacerbated this issue drastically. False information spreads incredibly quickly through various platforms, such as; news outlets, blogs, and social media networks. This allows false information to reach millions (if not billions) of people and influence the mainstream public opinions on topics such as, health, politics and social issues (Denniss & Lindberg, 2025: Muhammed & Mathew, 2022). Manual detection methods are not scalable to the magnitude of information being pushed out into the social zeitgeist, nor can it keep up with the constant evolution of the digital misinformation being spread.

X is a massive spreader of misinformation, (Suarez-Lledo & Alvarez-Galvez, 2021), it is one of the largest social media sites with 557 million people(this stat refers to potential advertising reach and not monthly users, there is a chance that much more people use X.com than stated here, (Biggest social media platforms by users 2025)). This is why it is critical to focus on it, the site has a massive sway on the public opinions of its users, and as it has such a large global user base I think focusing on it is the best choice.

The key challenges are in identifying and flagging misinformation at scale, in real-time, with a wide array of languages and regions having a diversity of experiences, multimedia formats and the constantly evolving tactics and forms of information being distributed. This project aims to address some of these difficulties, such as; feature extraction from short-form messages that are posted publicly, detection of multimodal (text + video/images) misinformation and minimising the errors that might occur due to dataset biases, ((Hamed et al., 2023; Xiao et al., 2025).

- **Significance**:

  Social trust is important and we cannot have a cohesive and secure society without combating misinformation on X. It is vital for digital safeguarding, supporting public health and preserving the quality of the informational environment. Creating a form of automated detection will be of great assistance to fact-checkers, platform moderators and policymakers in responding to quick misinformation surges, this is most important during health crises and elections (Suarez-Lledo and Alvarez-Galvez, 2021). This project will help find answers and create a stepping stone to the growing need for scalable, ethical and effective digital misinformation control in social media.

2. Aims, Objectives, and Scope

- **Aim**:

  The primary objective of this project is to develop and deploy an ML-based misinformation detection application for X that can assist in identifying, classifying and flagging potentially misleading or false posts. The project will aim to incorporate multimodal analysis and use innovative, pre-trained models (XFacta; Xiao et al., 2025).

- **Objectives**:
  - Enable the users to paste a link to the X post and automatically fetch its full content.
  - Extract and process the multimodal information on the post linked and analysing it.
  - Analyse the metadata of the post for deeper context and potential propagation analysis.
  - Detect and flag potentially misleading, false, or harmful information in the post using ML-powered models.
  - Display clear misinformation risk score and explanation to the user.
  - Ensure user privacy and data security when analysing external posts.
- **Scope**:
  - **Inclusions**:
    - Area to post links to X posts.

    - Easy to understand risk scoring system for the analysed posts.
    - Real-time scanning of posts with multimodal analysis.
    - Generated report for all the X posts being analysed by the system.
    - Model retraining using new misinformation datasets, training the model to be able to analyse new patterns and tactics.

  - **Exclusions**:
    - Direct platform moderation (this is more of an advisory tool and assistive tool, it is not being produced in collaboration with X).
    - Direct Message (DM) detection (due to privacy constraints and to help maintain better security).
    - Direct way for users to post feedback (not a functionality that I view as important in this iteration of the product/project).
    - Any user authentication or account holding system (there is no data that I view as important to be stored, also similar security and privacy concerns as the first point).
    - None web based features (the entire project requires direct communications with the X API and has many more vital web based features).
    - Separate application instance that does not function in a browser (same reason as above point, as well as no need to download anything that is being used by the system).

3. Background Review

- **Key Findings (Literature / Systems)**:
  - Health misinformation spreads incredibly quickly and over a large space (globally) on platforms like X, this creates a big threat to the public health of the entire population, (Suarez-Lledo & Alvarez-Galvez, 2021; Denniss & Lindberg, 2025).

- Social media sites have incredibly huge user bases (especially one like X (Statista, 2025), they greatly amplify the reach and impact of false information being spread.
- Classical ML methods (e.g., SVM, decision trees) are the foundation for fake news detection but are not sophisticated enough to handle the constantly evolving misinformation strategies, (Aphiwongsophon & Chongstitvatana, 2018).
- Significant challenges that persist throughout this line of work and its methodology: biased datasets, feature engineering limits and need for integrating multimodal (text, images, networks) data, (Hamed et al., 2023).
- There are huge advances in machine learning that can greatly improve misinformation detection, such as, improvements in transformer architecture, improvements in language models and improvements in transfer learning (Hu et al., 2025; Alshuwaier & Alsulaiman, 2025).
- In 2025 we now have cutting-edge approaches benchmarked by XFacta, through the use of diverse datasets and integrated analysis (Xiao et al., 2025).

- **Gaps Identified in Current Research/Technology**:
  - Most solutions are only text based, the lack of multimodal analysis is a flaw as it neglects the extra context that can be found in images and videos.
  - There is a lack in real-time analysis as well as no valuable explainability in the analysis that is being done on the post.
  - Continual model updating and retaining are rarely implemented, this will reduce how adaptable the model is to the evolving tactics of misinformation.

- **Comparison with Similar Software Applications/Products**
  - **Classical ML-based solutions:**
    - **Pros:** Have a useful baseline, use well-understood and easy to learn algorithms.
    - **Cons:** Limited to only text, have static analysis, lack real-time capabilities and adaptability.
  - **Rule-based/Crowd-sourced moderation tools:**
    - **Pros:** Human insight and knowledge is better overall as we have the adaptability and skills to keep consistent with modern news cycles and fact check on the fly, flexible rule creation and adaption.
    - **Cons:** Slow, not scalable, lack of multimedia support

- **Platforms with batch or backend analysis:**
    - **Pros:** Handles large data, vital for integrating models like I will be
    - **Cons:** Results and data can be outdated, minimal support for real-time and multimodal input.

- **Differentiation**:
    - Will allow the users to paste a link to any X post and automatically fetch and analyse the text and images on the post for misinformation detection.
    - Will use multimodal LLMs (e.g., XFacta-benchmarked models) for analysis of the posts.
    - Will have a real time scoring and classification system, with short user-friendly explanations.
    - Will be deigned with support for easy re-training and model upgrades, as misinformation trends tend to easily evolve.

Feature/Capability	Classical ML Solutions	Existing Social Media Tools	XFacta (2025)	My Project
Text Analysis	**Yes**	**Yes**	**Yes**	**Yes**
Image/Multimodal Support	**No**	**Rarely**	**Yes**	**Yes**
Real-time User Interaction	**No/Limited**	**Often No**	**Possible**	**Yes**
Misinformation Scoring/Confidence	**Basic**	**Limited**	**Strong**	**Yes (strong, per XFacta)**
Use of Latest LLMs/Transformers	**No**	**Some platforms**	**Yes**	**Yes**

User Link Submission	**No**	**No**	**No**	**Yes (simple paste-to-analyse)**
Explainability	**No/Limited**	**Limited**	**Yes**	**Yes**
Dataset/Model Updating	**Rare**	**Rarely**	**Built-in**	**Built-in**

4. Tools (Hardware/Software)

- **Hardware**:
  - Laptop/PC
- **Software**:
  - **Python:** My model training and backend will be written in this language, I will use plenty of libraries and frameworks that are linked to python (pip, FastAPI, Jupyter notebook and PyTorch).
  - **React or Vue.js:** This will be used to create the frontend and interactivity of what the user sees/uses with the backend
  - **FastAPI:** This will be my framework of choice for the backend. I chose it due to it being very lightweight, having incredibly good API functionality which I will need for analysing the X posts as well as being the primary source of backend communication. I can use it to create a fast and efficient API for my project.
  - **VS Code:** This is where I will prototype, test, and document all my code and my model training. All of my none model training will also be done here, this is because it is quite easy to use and has a vast amount of support for everything I need and I can just use it for everything instead of going between different code editors/IDEs
  - **Jupyter Notebooks (in VS Code):** This a very easy to use and accessible tool to use for data science and the training of machine learning models, there are even other versions of it like google collab. I am using it inside VS Code because it just makes my life easier if everything is done in one place rather than scattered around and also the alternatives are not better enough to warrant me using them.

- **X(Twitter) API v2:** This will be needed to fetch the linked X posts and their metadata that the model will analyse
- **Transformers (Hugging Face Library):** This is how I will access pre-trained models that I will evaluate (e.g., XFacta)
- **GIT/GITHUB:** This is how I will maintain version control and have the ability to track any changes I make, so in the unfortunate chance I make a huge error I can go back to a version that is more stable and continue from there.
- **TensorBoard or MLFlow:** This will be used to evaluate my model vs other models like XFacta and make sure that it is at a high enough standard to be used.

5. Use Cases and Diagrams

- **Use Cases**:
  - **Checking Misinformation for Specific Post:**

The user will paste a link to an X post. The system will then fetch all available text, images and metadata from the post, it will process all the gathered information and content, analyse them and then display a misinformation risk score with an explanation.

  - **Batch Analysis of Multiple Posts:**

Multiple X posts can be pasted for batch analysis. The system will sequentially analyse them, return risk scores and provide a downloadable report.

  - **Viewing of Analysis Report:**

After the analysis of an X post, the user views a report summarising what was analysed, its risk score and ana explanation. This report will also be downloadable.

  - **Model Review:**

This is similar to the report, users will be able to see factors (e.g., words, images/text mismatch and source reliability) that contributed to the results of the report. This will be added to the report.

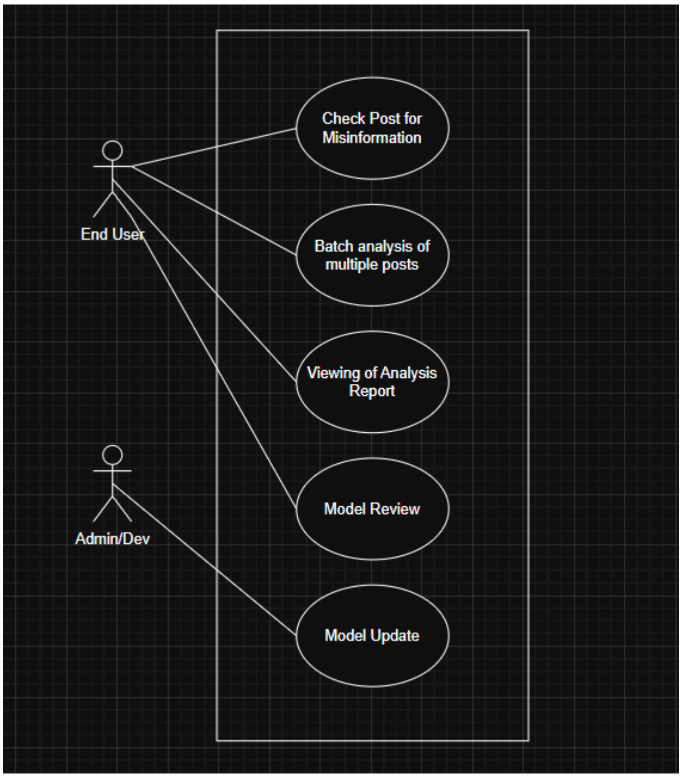
  - **Model Update (Developer):**

The system owner will be able to update core ML models and retrain them using new datasets or samples, this helps keep the system adaptable to the ever changing tactics being used. It also allows the system to maintain real-time ability to combat misinformation.

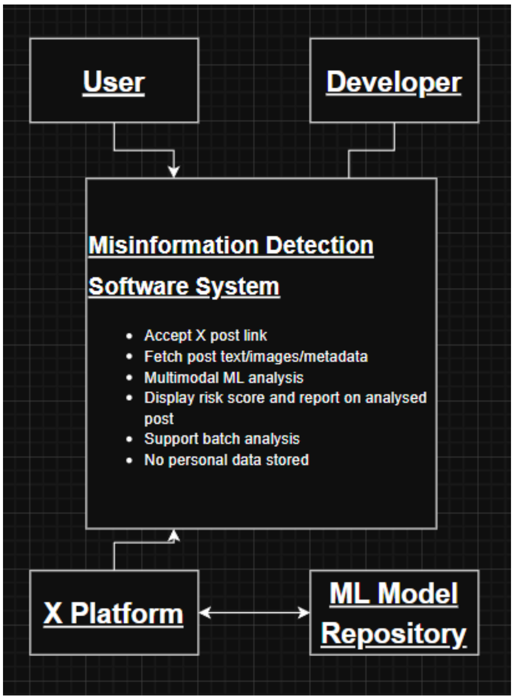
- **Ensure Data Security:**

  The system will not have any way of storing user information, all data will be processed securely, without storing user results beyond each session

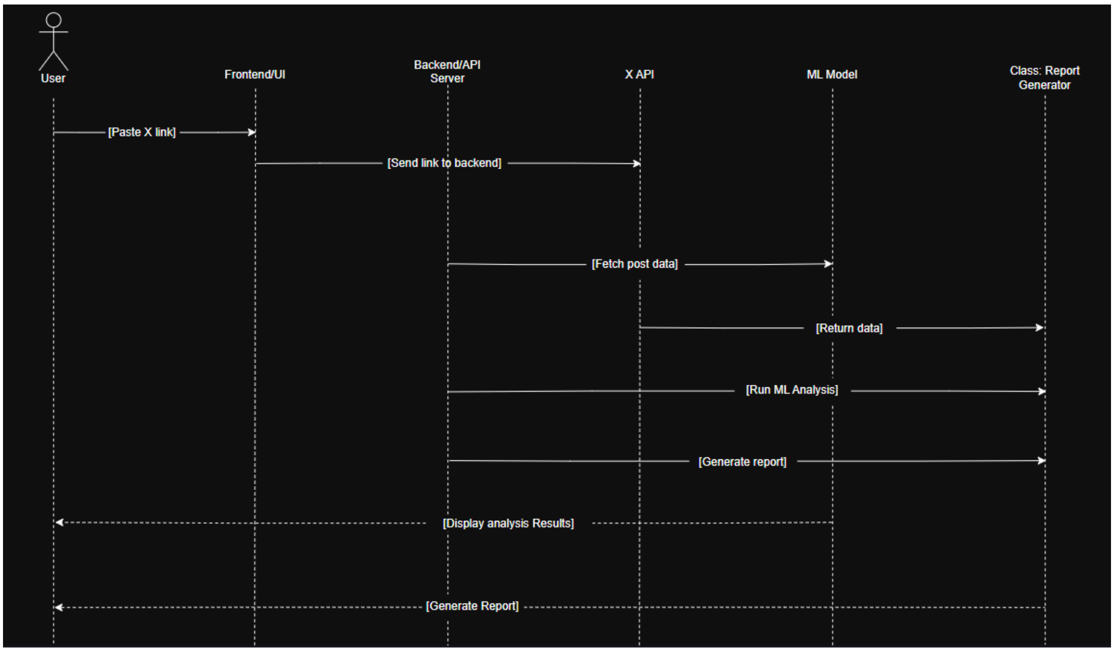
- **Diagrams**:
  - **Use Case Diagram: [Link]**



  - **System Context Diagram: [Link]**



- **Sequence Diagram: [Link]**



6. Requirements Elicitation

- **Approach**:
  - **[Method 1]**: **Literature and System Analysis**

I will analyse recent academic studies and similar existing tools for functional patterns, gaps and user needs that are relevant and specific to my project.

  - **[Method 2]**: **Prototyping and Self-use Testing**

I will develop and iteratively refine the model and overall application/software, this will be done by going through use cases, identifying additional requirements and usability improvements, this will all be done by me in a very hands-on and technical approach. I will use this as an opportunity to refine the functionality of my model and to improve its abilities.

- **Techniques**:
  - Desk-based research. This is reviewing academic publications, benchmarking features and making sure that my system evaluation metrics are the best I can make them.
  - Reverse engineering of public online misinformation products for workflow ideas and analysis optimisation.

- Prototyping and evaluating the project using Jupyter notebooks and frontend mockups.
- Requirements documentation with structured tables/lists to track against the implementation that I have done.

- **Requirement Specification**
  - **Functional Requirements:**
    - The system must allow users to paste a URL of any public X post.
    - The system must fetch the text and images/video content for each linked post via the X API.
    - The system must be able to process and combine all fetched data for analysis by a multimodal ML model.
    - The system must classify posts as "Likely True", "Likely Misinformation", or "Uncertain"; it must also show a risk/confidence score for said classification.
    - The system must display an explanation of the factors that are contributing to the model's decision and why it views them as key to its evaluation and results.
    - The system must provide the user a summary report after each scan and also provide an option to download the report.
    - The system must provide the ability to analyse multiple links at once (batch analysis, the user can paste multiple links at once).
    - The system must allow for model retraining using newly gathered labeled data.
    - The system must ensure that all user input and analysis results are processed securely and are deleted after a session ends, no information from previous sessions by any users is to be stored.
  - **Non-Functional Requirements**
    - System operations must be completed in real-time (an acceptable range for response time should be 10 ~ 15 seconds per link posted).
    - The user interface should be easy to read and follow for all users, it should provide its information in a way that does not confuse anyone.

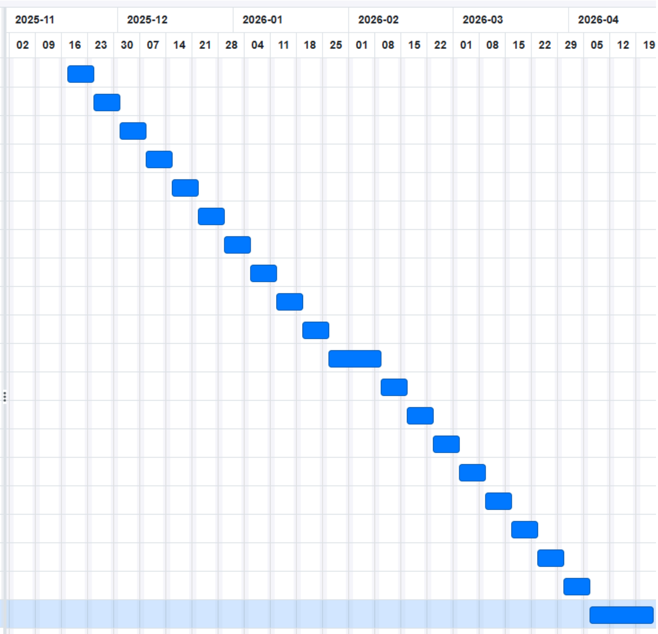
- The system must provide clear error messaging for any input, retrieval or analysis failure that might occur.
- All sensitive information must be handled in accordance with data privacy standards, with no user identifiable information being permanently stored anywhere on the system.
- The software must be created in a way where future extensions can be added (e.g., support for other platforms or new ML models).

7. Time Schedule

- **Gantt Chart Overview**:
  - Nov 17-23, 2025: More background research, review PPRS feedback
  - Nov 24-30, 2025: Literature review, dataset collection, system architecture planning
  - Dec 1-7, 2025: Requirement details finalisation, finalise tech stack decision, Github creation and code base prep
  - Dec 8-14, 2025: Double check requirements, initial data cleaning
  - Dec 15-21, 2025: Data preparation and baseline traditional models training
  - Dec 22-28, 2025: Model evaluation and tuning, start UI design
  - Dec 29-Jan 4, 2026: Prototyping key ML components, REST API planning
  - Jan 5-11, 2026: Finalised ML training and fine tuning (with core features)
  - Jan 12-18, 2026: Prototype web/API integration, UI development
  - Jan 19-25, 2026: Core integration of frontend, backend and API's, error handling, documentation drafting
  - Jan 26-Feb 1, 2026: Polishing code, IPD prep with supervisor
  - Feb 2-8, 2026: Interim Progress Demo (Assessment: Feb 5)
  - Feb 9-15, 2026: Use given feedback and adapt on it, expand UX, improve model performance
  - Feb 16-22, 2026: Advanced feature implementation, testing, validation
  - Feb 23-29, 2026: Code refinements, (testing, addition of unadded features and validation)
  - Mar 1-7, 2026: Deployment, bug fixing
  - Mar 8-14, 2026: User and system testing, performance improvement

- o Mar 15-21, 2026: Draft final report sections: Methods, System Design
- o Mar 22-28, 2026: Finalize implementation, write Results section, diagrams
- o Mar 29-Apr 4, 2026: Full system debug, review, demo video prep
- o Apr 5-11, 2026: Edit final report, system review, viva prep
- o Apr 12-18, 2026: Supervisor review, last edits, more viva prep
- o Apr 19-23, 2026: Submit Final Project & oral viva (Assessment: Apr 23)

ID	Task Name	Start	End	Duration
1	More background research, review PPRS feedback	2025-11-17	2025-11-24	6 days
2	Literature review, dataset collection, system architecture planning	2025-11-24	2025-12-01	6 days
3	Requirement details finalisation, finalise tech stack decision, Github creation and code base prep	2025-12-01	2025-12-08	6 days
4	Double check requirements, initial data cleaning	2025-12-08	2025-12-15	6 days
5	Data preparation and baseline traditional models training	2025-12-15	2025-12-22	6 days
6	Model evaluation and tuning, start UI design	2025-12-22	2025-12-29	6 days
7	Prototyping key ML components, REST API planning	2025-12-29	2026-01-05	6 days
8	Finalised ML training and fine tuning (with core features)	2026-01-05	2026-01-12	6 days
9	Prototype web/API integration, UI development	2026-01-12	2026-01-19	6 days
10	Core integration of frontend, backend and API's, error handling, documentation drafting	2026-01-19	2026-01-26	6 days
11	Polishing code, IPD prep with supervisor & Interim Progress Demo (Assessment: Feb 5)	2026-01-26	2026-02-09	11 days
12	Use given feedback and adapt on it, expand UX, improve model performance	2026-02-09	2026-02-16	6 days
13	Advanced feature implementation, testing, validation	2026-02-16	2026-02-23	6 days
14	Code refinements, (testing, addition of unadded features and validation)	2026-02-23	2026-03-02	6 days
15	Deployment, bug fixing	2026-03-02	2026-03-09	6 days
16	User and system testing, performance improvement	2026-03-09	2026-03-16	6 days
17	Draft final report sections: Methods, System Design	2026-03-16	2026-03-23	6 days
18	Finalize implementation, write Results section, diagrams	2026-03-23	2026-03-30	6 days
19	Full system debug, review, demo video prep	2026-03-30	2026-04-06	6 days
20	Final report, viva prep and project submission	2026-04-06	2026-04-23	14 days



8. References / Bibliography

- Suarez-Lledo, V. and Alvarez-Galvez, J. (2021). Prevalence of Health Misinformation on Social Media: Systematic Review. *Journal of Medical Internet Research*, 23 (1), e17187. Available from https://doi.org/10.2196/17187.

- Hamed, S.K., Ab Aziz, M.J. and Yaakub, M.R. (2023). A review of fake news detection approaches: A critical analysis of relevant studies and highlighting key challenges associated with the dataset, feature representation, and data fusion. *Heliyon*, 9 (10), e20382. Available from https://doi.org/10.1016/j.heliyon.2023.e20382.

- Hu, B., Mao, Z. and Zhang, Y. (2025). An overview of fake news detection: From a new perspective. *Fundamental Research*, 5 (1), 332–346. Available from https://doi.org/10.1016/j.fmre.2024.01.017.
- Aphiwongsophon, S. and Chongstitvatana, P. (2018). Detecting Fake News with Machine Learning Method. *2018 15th International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON)*. July 2018. 528–531. Available from https://doi.org/10.1109/ECTICon.2018.8620051 [Accessed 15 October 2025].
- Alshuwaier, F.A. and Alsulaiman, F.A. (2025). Fake News Detection Using Machine Learning and Deep Learning Algorithms: A Comprehensive Review and Future Perspectives. *Computers*, 14 (9), 394. Available from https://doi.org/10.3390/computers14090394.
- Denniss, E. and Lindberg, R. (2025). Social media and the spread of misinformation: infectious and a threat to public health. *Health Promotion International*, 40 (2), daaf023. Available from https://doi.org/10.1093/heapro/daaf023.
- Muhammed T, S. and Mathew, S.K. (2022). The disaster of misinformation: a review of research in social media. *International Journal of Data Science and Analytics*, 13 (4), 271–285. Available from https://doi.org/10.1007/s41060-022-00311-6.
- Biggest social media platforms by users 2025. (no date). *Statista*. Available from https://www.statista.com/statistics/272014/global-social-networks-ranked-by-number-of-users/ [Accessed 3 November 2025].
- Xiao, Y. et al. (2025). XFacta: Contemporary, Real-World Dataset and Evaluation for Multimodal Misinformation Detection with Multimodal LLMs. Available from https://doi.org/10.48550/ARXIV.2508.09999 [Accessed 3 November 2025].

9. Ethics Form